Rarefied hypersonic cylinder

Operator learning, baselines, and honest UQ

From author-supplied DSMC fields to a leakage-controlled teaching experiment

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Companion: notebooks/week07_1/W7_1_Hypersonic_Rarefied_Cylinder_DeepONet.ipynb

What this increment adds

OUTCOMES

- Separate freestream Mach from the predicted local Mach field.
- Formulate a parameter-to-field operator for DSMC cylinder data.
- Freeze whole-case splits before model selection.
- Read the branch, trunk, layerwise fusion, and output head.
- Challenge the neural model with a structured interpolation baseline.
- Audit training error; a single MLP does not provide calibrated uncertainty.

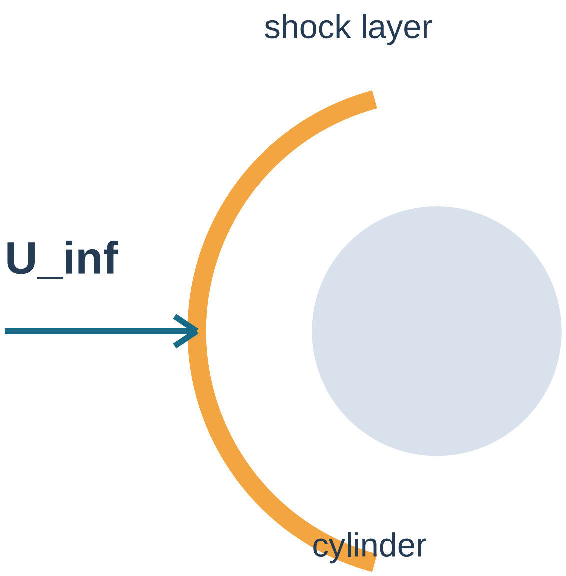
CLAIM

The lab executes a compact, author-released DSMC derivative. The default CPU model is a **teaching analog**; it does not reproduce the paper's full-resolution accuracy or speedup.

RULE

If a transparent baseline wins, retain that result. Complexity is not evidence.

Physics: three scales, three questions



Mach number: $M_{inf} = U_{inf}/a_{inf}$ controls compressibility. The target $M(x,y)$ is a different, spatial field.

Knudsen number: Kn = lambda/D measures rarefaction. As lambda becomes non-negligible, continuum closure and no-slip assumptions need qualification.

Sampling scale: DSMC fields estimate particle moments. A sharp shock and finite particle count make target noise spatially nonuniform.

Mach independence and the classroom MLP

Physical motivation. At high Mach, appropriately nondimensionalized fields can vary slowly with Mach under compatible gas, rarefaction, geometry and boundary conditions. This motivates interpolation; a small error alone is not proof of Mach independence. The source TOV/P scaling must be verified.

Required model. Three 96-unit tanh layers; inputs (Mach, x, y); equal standardized-target MSE. Scalers use training cases only. Adam, fixed seed 760, maximum 300 epochs and 40-epoch patience; best epoch selected on four whole validation cases. Report training error as an underfitting diagnostic.

Provenance. Research DSMC data precede the course. This MLP is a new teaching model, not the paper's trained Fusion-DeepONet. A single fit provides no calibrated UQ. Broader data licensing awaits rights-holder approval.

The learning object is an operator

Branch input
freestream M_inf

→

G_theta: M_inf -> q(x,y)

q = [local Mach, source TOV, source P]

→

Query
(x,y) -> q

A field surrogate must be judged at unseen flow cases, not merely unseen grid points from a case already used during training.

Data audit: small derivative, complete provenance

20	**400 x 400**	**50 x 50**	**44,500**	**0.5 MB**
Mach cases	source grid per case	deterministic selection	finite retained points	committed NPZ

KEPT

- M_inf, x, y
- local Mach, source TOV, source P (legacy ratio keys do not verify scaling)
- case ID and original source-grid row
- archive and derivative SHA-256 hashes

EXCLUDED

- 1.4 GB source ZIP and 4.9 GB expanded files
- training logs and duplicate checkpoints
- historical scripts with inconsistent protocols
- research checkpoints not needed for the lab

Whole-case splitting prevents a false claim

TRAIN	VALIDATION	BLIND INTERP.	BLIND EXTRA.
5, 6, 7, 8, 9, 10, 11, 12, 13, 14	8.25, 8.75, 9.25, 9.75	5.5, 6.5, 7.5, 8.5, 9.5	15

Wrong question: can the network fill random points from a Mach case it has already seen?
Scientific question: can it predict a complete field at an operating condition excluded from fitting and tuning?

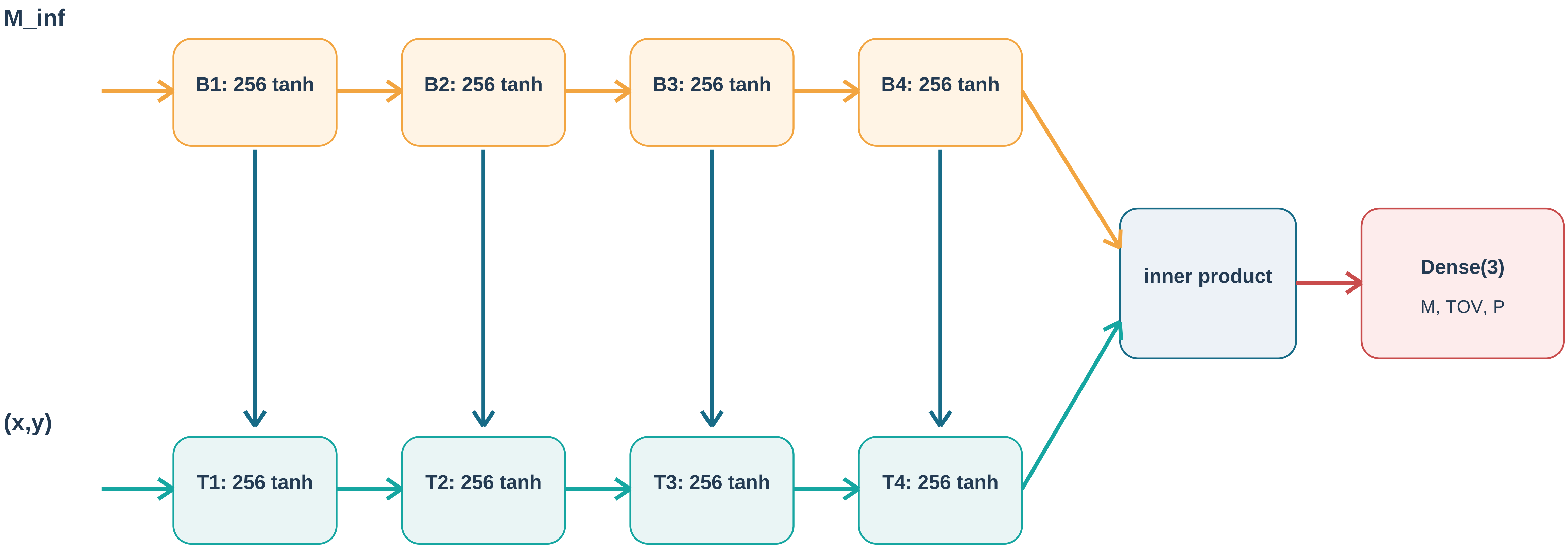
The mandatory baseline is field interpolation

$$\hat{q}(M_q) = q(M_a) + \left[\frac{M_q - M_a}{M_b - M_a}\right]\left[q(M_b) - q(M_a)\right]$$

Split	Local Mach	Temperature	Pressure
Blind interpolation	**0.398%**	**0.609%**	**0.779%**
Mach 15 extrapolation	0.538%	0.877%	0.848%

Result: a neural operator must beat a sub-1% structured baseline here, or justify itself through a different deployment constraint.

Reviewed Fusion-DeepONet topology



Each block includes dropout in the reviewed stored model. Branch state is added to the matching trunk depth before the final branch-trunk combination.

Training protocol: every choice changes the claim

1. **Fit scalers** — Training cases only; MinMax inputs and standardized targets.
2. **Optimize** — Adam with validation-monitored early stopping and learning-rate reduction.
3. **Weight** — Reviewed archive loss: $MSE_M + MSE_T + 5\ MSE_p$ in standardized space.
4. **Repeat** — Five independent initializations; preserve member predictions.
5. **Open blind cases** — After design is frozen; report interpolation and extrapolation separately.

Optional ensembles: not run by the single-MLP lab

Five fits
different initial weights
same frozen protocol

→

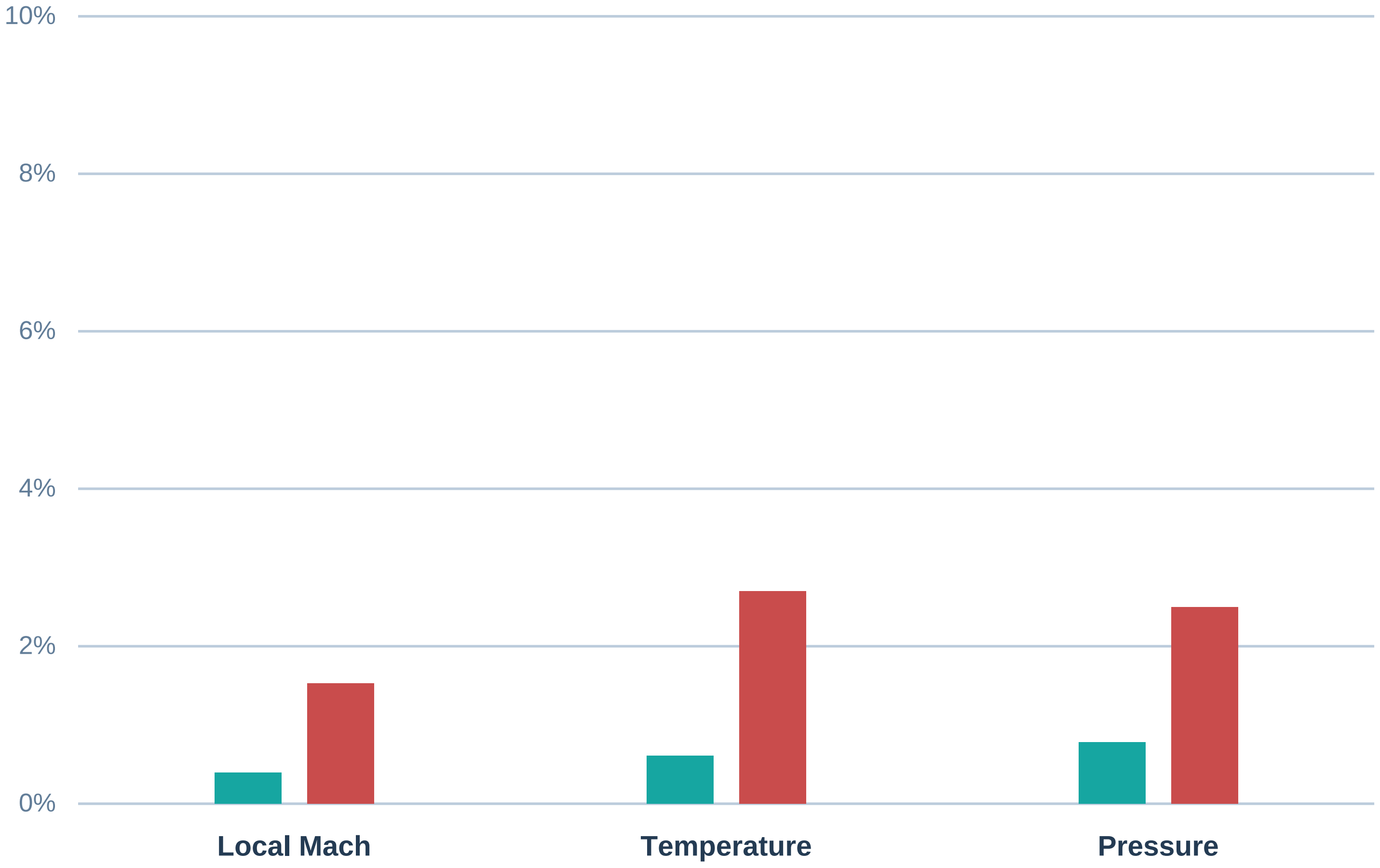
At every point
mean mu(x,y)
sample spread s(x,y)

→

Audit
Does truth fall in mu +/- 2s?

Low coverage means overconfidence. High coverage can also be uninformative if intervals are too wide. Report coverage and interval width, by target and by split.

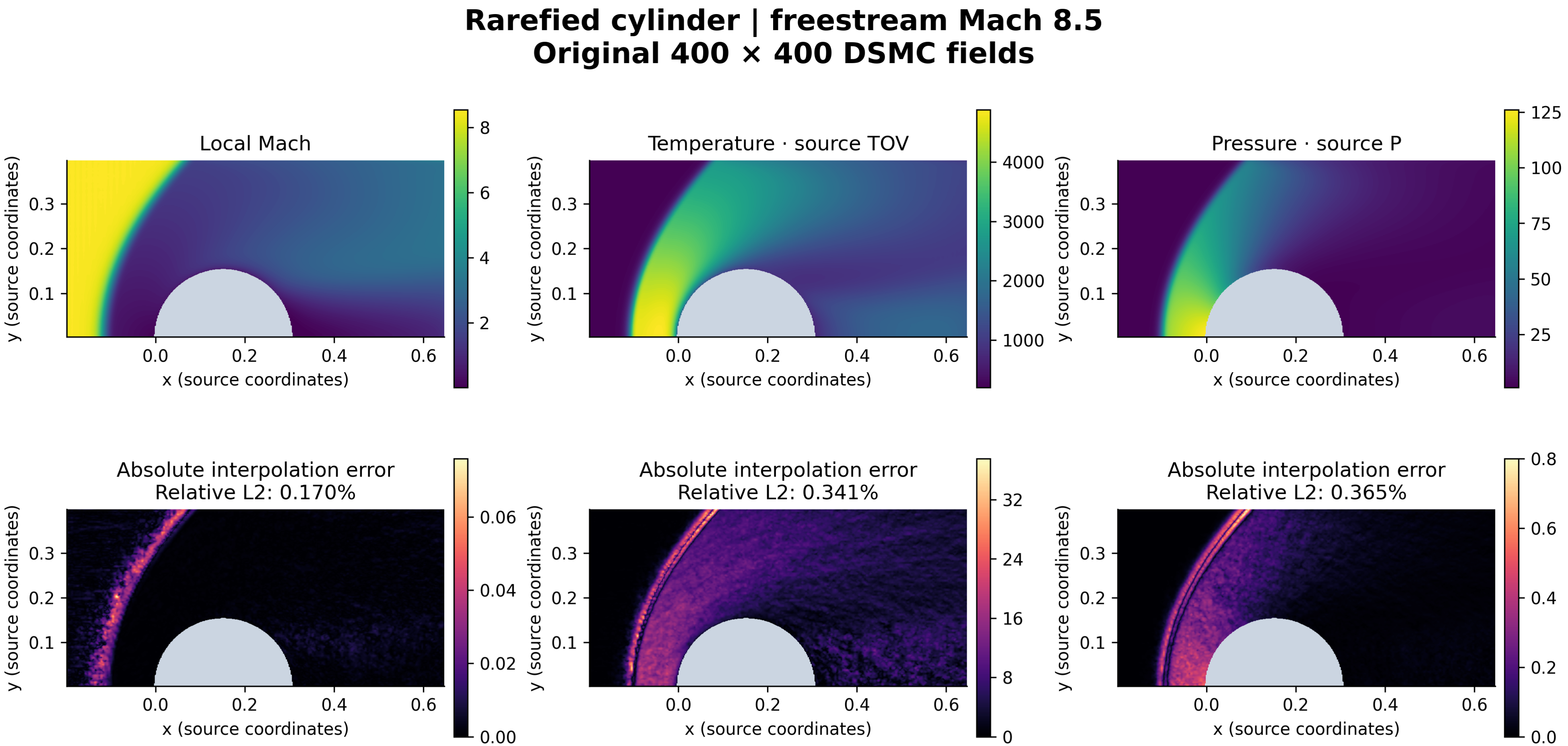
Executed evidence: trained MLP versus interpolation



Baseline
0.398% | 0.609% | 0.779%

Trained 3x96 MLP
1.53% | 2.70% | 2.50%

Full source-grid interpolation: separate from MLP metrics



Gray: masked solid/sentinel region. Interpolation uses Mach 8 and 9; no spatial smoothing.

Decision gate: when is a neural operator justified?

Accuracy
Does it beat field interpolation on frozen blind cases?

Geometry
Must one model span grids, shapes, or boundary conditions?

Availability
Will deployment lack bracketing high-fidelity solutions?

Cost
Do many field queries amortize training and data production?

Physics
Are shock location, surface loads, positivity, and limits acceptable?

Uncertainty
Is spread calibrated on the deployment regime?

Lab deliverables and claim boundary

SUBMIT

- Baseline and teaching-operator relative-L2 errors by target and split.
- One physics-based interpretation of an error hot spot.
- Training error and an explicit statement that the single MLP has no calibrated UQ.
- A short explanation of random-point leakage.
- One deployment condition that could justify full neural training.

MAY CLAIM

The compact DSMC derivative, frozen split, baseline, and CPU teaching analog were executed.

MAY NOT CLAIM

Published full-model accuracy, speedup, complete-field reproduction, or calibrated UQ.

WEEK 7.1

References and reproducibility trail

- Roohi et al. (2026), Neural Networks for Rarefied Gas Dynamics: Relaxation Problem, Polyatomic Shock Waves, and Hypersonic Cylinder Flow. Physics of Fluids 38, 057108. https://doi.org/10.1063/5.0334590
- Lu et al. (2021), Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. Nature Machine Intelligence 3, 218-229. https://doi.org/10.1038/s42256-021-00302-5
- Lakshminarayanan, Pritzel, and Blundell (2017), Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles. NeurIPS 30.
- Bird (1994), Molecular Gas Dynamics and the Direct Simulation of Gas Flows. Oxford University Press.

Rebuild data: qa/build_hypersonic_cylinder_subset.py | Rebuild evidence: qa/run_hypersonic_cylinder_evidence.py | Source archive SHA-256 begins bda221759a37